American Computer Science League

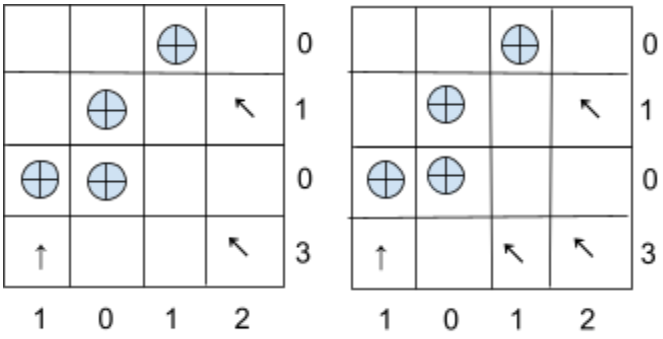
2022-2023 ● Contest 4: Targets ● Senior Division

PROBLEM: You are given an *n* x *n* grid, some target locations in that grid, border numbers that indicate how many arrows appear in each row and column, and the locations of some arrows. Each arrow points to one and only one target. One arrow will be missing. Find the location and the direction of the missing arrow. The directions of the arrows are: **(A) ←, (B) ↑, (C) →, (D) ↓, (E) ↖, (F) ↗, (G) ↘,** and **(H)** ↙. An arrow cannot point toward another arrow before reaching its target. No cell can have more than one arrow. Cells will be labeled using (row, column) format with (0,0) in the upper left-hand corner using row-major order.

EXAMPLE: Given the following input:

```
4
02 11 20 21
0103 1012
13E 30B 33E
```

It will be a 4 x 4 grid with targets in cells (0,2), (1,1), (2,0), and (2,1). The border numbers will be 0, 1, 0, and 3 from top to bottom and 1, 0, 1, and 2 from left to right. The arrows 13E, 30B, and 33E are given. Since the target at cell (2,1) is not pointed to by the given arrows, the arrow needed is at cell (3,2) with direction E so the output is 32E so that the column and row border numbers giving the number of arrows are also correct. We guarantee a unique solution.



INPUT: The input will consist of 4 lines of data:

- a number, *size*, for the size of the grid
- a string, *targets*, for the locations of the targets as 2-character strings, each separated by a single space
- a string, *numbers*, for the number of arrows in each row (on the right-side of the grid) followed by the number of arrows in each column (on the bottom of the grid) separated by a single space
- a string, *arrows,* for the locations and directions of the given arrows as 3-character strings, each separated by a single space.

We guarantee that the grid will be no larger than 10 x 10.

OUTPUT: For each grid, output the 3-character string representing the location where the arrow must be placed and the direction of the arrow.

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SAMPLE INPUT:

```
1. 4
   02 11 20 21
   0103 1012
   13E 30B 33E
2. 6
   15 23 24 32 33 34 42 43 51
   401211 401211
   00G 20G 40G 53B 02G 03D 04G 35A
3. 6
   00 10 13 20 21 24 30 31 43 50 53
   420113 022124
   01H 02H 03H 04H 15H 14H 45A 51B 55E 52E
4. 6
   02 05 13 23 12 35 24 00
   011114 410111
   14F 20F 50F 53F 51F 40F 30F
5. 6
   01 10 23 42 53 12 04 52 00
   200232 111024
   05H 34H 35H 54E 55E 40F 45A 41B
```

SAMPLE OUTPUT:

```
1. 32E
2. 30C
3. 35H
4. 55E
5. 02D
```

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TEST DATA

TEST INPUT:

1. 6
   02 04 10 13 20 33 35 40 44 55
   202204 212122
   00C 05A 22F 25D 32E 50B 51F 53E 54C
2. 6
   01 10 20 21 22 23 24 31 45 51
   310123 100441
   03A 04D 05H 13A 34A 43E 44E 53A 54F
3. 6
   11 12 13 30 35 41 42 43 45 51 52
   323102 422102
   00G 01G 02G 10G 15D 20D 22D 23G 31D 50C
4. 6
   11 20 22 31 32 33 40 42 44 50 51 53
   411132 231114
   00D 01D 04H 05H 15H 21D 30D 43A 45A 52A 55A
5. 6
   05 11 20 21 22 23 33 42 43 52 53 54
   321312 401043
   00D 02H 04H 14H 15B 24H 34H 35H 40C 50C 55A

TEST OUTPUT:

1. 34D
2. 50B
3. 55A
4. 41H
5. 30F

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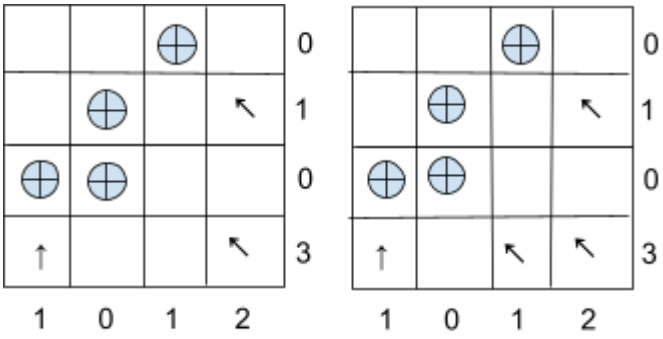
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问题: 给定一个 $n \times n$ **方格表，其**中**有**一些目标位置、**表**示每行和每列中箭头**数量**的**数字**以及一些箭头的位置。每个箭头指向**有且只有**一个目标。**方格表中**缺少了一个箭头。**请**找到缺失箭头的位置和方向。箭头的方向是：**(A)** ←、 **(B)** ↑、**(C)** →、**(D)**↓、 **(E)** ↖、**(F)** ↗、**(G)** ↘ 和 **(H)** ↙。一支箭**头**在到达目标之前不能指向另一支箭**头**。任何单元格内的**箭头**都不能**超过一**个。单元格将**按照**（行，列）**的**格式进行标记，左上角**以行优先的方式标记为**（0, 0）。

示例: 给**定以**下输入：

```
4
02 11 20 21
0103 1012
13E 30B 33E
```

它将是一个 4 x 4 方格表，其中目标位于单元格（0, 2）、（1, 1）、（2, 0）和（2, 1）中。方格表边界上的数字，从上到下依次为 0、1、0 和 3，从左到右依次为 1、0、1 和 2。给出箭头 13E、30B 和 33E。由于给定箭头没有指向单元格（2, 1）处的目标，因此所需的箭头位于单元格（3, 2）方向为 E，输出为 32E，这样表明箭头数量的行列边界数字也是正确的。我们保证仅有唯一答案。



输入: 输入将由 4 行数据组成：

- 一个数字表示大小，即方格表的大小
- 一个字符串表示目标，即将目标的位置表示为 2 个字符的字符串，每个字符串之间由一个空格进行分隔
- 一个字符串，表示数量，即表示每行（在方格表的右侧）的箭头数量，紧接着每列（在方格表底部）的箭头数量，中间由一个空格分隔。
- 一个字符串，表示箭头，即用有 3 个字符的字符串表示给定箭头的位置和方向，每个字符串之间由一个空格分隔。

  我们保证方格表的大小不会大于 10 × 10。

输出: 对于每个方格表，输出有 3 个字符的字符串，表示必须放置箭头的位置和箭头的方向。

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样本输入:

```
1. 4
   02 11 20 21
   0103 1012
   13E 30B 33E
2. 6
   15 23 24 32 33 34 42 43 51
   401211 401211
   00G 20G 40G 53B 02G 03D 04G 35A
3. 6
   00 10 13 20 21 24 30 31 43 50 53
   420113 022124
   01H 02H 03H 04H 15H 14H 45A 51B 55E 52E
4. 6
   02 05 13 23 12 35 24 00
   011114 410111
   14F 20F 50F 53F 51F 40F 30F
5. 6
   01 10 23 42 53 12 04 52 00
   200232 111024
   05H 34H 35H 54E 55E 40F 45A 41B
```

样本输出:

```
1. 32E
2. 30C
3. 35H
4. 55E
5. 02D
```

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测试数据

测试输入:

1. 6
   02 04 10 13 20 33 35 40 44 55
   202204 212122
   00C 05A 22F 25D 32E 50B 51F 53E 54C
2. 6
   01 10 20 21 22 23 24 31 45 51
   310123 100441
   03A 04D 05H 13A 34A 43E 44E 53A 54F
3. 6
   11 12 13 30 35 41 42 43 45 51 52
   323102 422102
   00G 01G 02G 10G 15D 20D 22D 23G 31D 50C
4. 6
   11 20 22 31 32 33 40 42 44 50 51 53
   411132 231114
   00D 01D 04H 05H 15H 21D 30D 43A 45A 52A 55A
5. 6
   05 11 20 21 22 23 33 42 43 52 53 54
   321312 401043
   00D 02H 04H 14H 15B 24H 34H 35H 40C 50C 55A

测试输出:

1. 34D
2. 50B
3. 55A
4. 41H
5. 30F